higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 4

18 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 C
1.2 B
1.3 A
1.4 C
1.5 D

(5 × 2) **[10]**

QUESTION 2

2.1 2.1.1 Cations
2.1.2 Cations are particles that are positively charged. (2 × 2) (4)

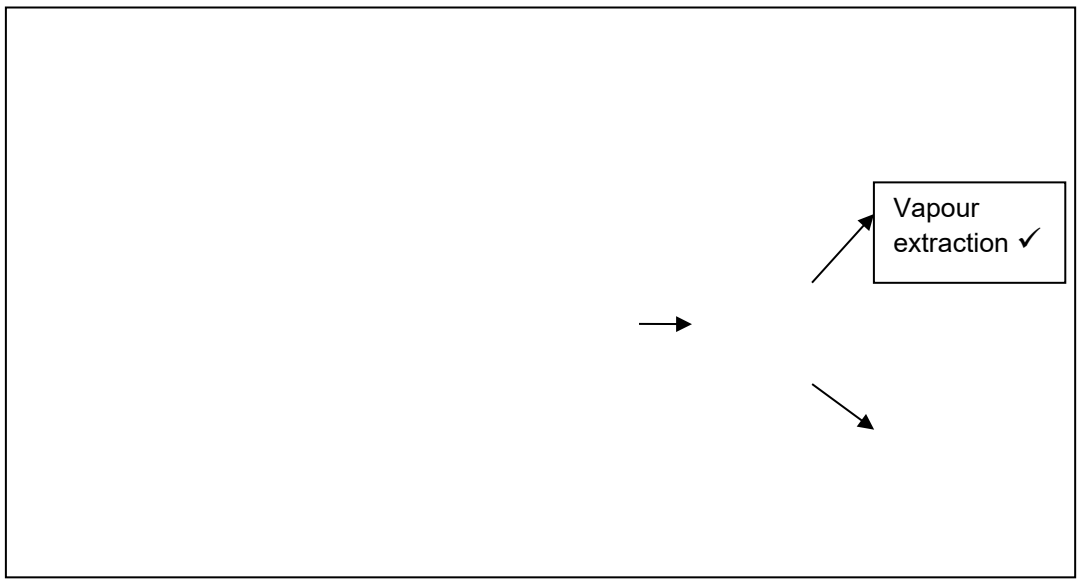
2.2 To form machine parts that are difficult to produce by any other method, like the manufacturing of valve bodies. (2)

2.3 It makes the galvanised steel resistant to corrosion. (2)

2.4
- Glue
- Silicon (2)

2.5
- Heat input phase
- Flashing and recovery phase
- Heat rejection phase (3)

2.6



(6 × 1) (6)

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2.7 Plate condensers operate like plate evaporators.✓ They have specialised plates stacked together,✓ each having a hole through which vapour and cooling liquid can flow.✓ They form a pipe-like system when stacked together.✓ The hot vapour enters at the top of the condenser.✓ As the vapour condenses, it flows to the bottom of the condenser through a piping system.✓ The piping system connects the top to the bottom section.✓ The condensed liquid is then removed from the condenser through the bottom pipes.✓ The cooling water enters the system at a low temperature.✓ As heat is transferred to it from vapour, it warms up and leaves at a higher temperature.✓ The vapour and cooling liquid are pumped into the condenser at specific speeds and pressures, depending on the design and cooling rate required.✓ (11)

2.8

VENTURI	CYCLONE
The gas and liquid enter through a pipe, having a smaller cross-section than the rest of the pipe. ✓	The gas enters the column at the circumference of the column. ✓
This increases the flow velocity. ✓	It then flows in a circular motion. ✓

(4)
[34]

QUESTION 3

3.1 3.1.1 When heat is transferred to an area which it is not meant for, heat loss occurs.

3.1.2 It is the temperature at which a liquid will change into a gas at atmospheric pressure.

(2 × 2) (4)

3.2
- Convection
- Conduction
- Radiation (3)

3.3 3.3.1 $E_K = ½ m v^2$

3.3.2 $E_P = m g h$

(2 × 3) (6)

3.4 When heat leaves a system it is negative,✓✓ but when heat is put into a system it is positive. ✓✓ (4)

3.5 $C = \frac{Q}{\Delta T}$ ✓

$Q = C \times \Delta T$ ✓

$= 0,39 \text{ kJ/K} \times 38,6$ ✓✓

$= 15,05 \text{ kJ}$ ✓✓

(6)

3.6 $Cv = \frac{Vp}{V}$ ✓

$= \frac{250\ litre}{10\ litre + 250\ litre}$ ✓

$= \frac{250\ litre}{260\ litre}$ ✓

= 0,962 ✓

= 9,62% ✓

(5)
[28]

QUESTION 4

4.1 Matter or mass cannot be created or destroyed; however, it can change from one form to another form. ✓✓✓
$m_{in} = m_{out}$ ✓✓ (5)

4.2 4.2.1 It is a fully defined space that is closed off to its surroundings by a boundary that prohibits any exchange of mass or energy.

4.2.2 It is the spatial dimension that defines a three-dimensional space in terms of cubic metres.

4.2.3 It is an indication of how many particles are present in a specified volume.

4.2.4 It is a process that involves the separation or combination of various atomic elements.

(4 × 2) (8)

4.3 It is the mass of one mole, given in grams per mole. (2)

4.4 Moles indicate the number of atomic particles present in a substance.✓ One mole of a substance contains the same number of particles given by Avogrado's number.✓ The Avogadro number is $6,022 \times 10^{23}$ particles.✓
Therefore, 1 mole of a substance or element contains $6,022 \times 10^{23}$ particles.✓
Correct answer (4)

4.5 $V = l \times b \times h$ ✓
$= 0{,}042\ m \times 0{,}017\ m \times 0{,}033\ m$ ✓
$= 0{,}0000235\ m^3$ ✓

$W = mg$ ✓
$= 0{,}0114\ kg \times 9{,}81\ m/s^2$ ✓
$= 0{,}112\ N$ ✓

$Y = \frac{W}{V}$ ✓

$= \frac{0{,}112\ N}{0{,}0000235\ m^3}$ ✓

$= 4\ 765{,}96\ N/m^3$ ✓

(9)
[28]

TOTAL: 100